

## **Problem 11.1 (MATLAB)**

### **Three-population classification using LDA and QDA**

The file '**dataset\_problem\_11\_1.mat**' contains size measurements for 150 observations of three species of iris (flowers). There are 50 observations of each species, coded in the first column of the data matrix,  $X$ , as "1", "2" and "3", respectively. The bivariate size data available in columns 2 and 3 is:  $x_1$  = sepal width [mm],  $x_2$  = petal width [mm].

Since definitive species determination of iris requires microscopic investigation, it is desired to investigate whether a simple discrimination might be possible, based solely on sepal and petal width measurement of a particular iris flower of unknown species.

Therefore: in this problem consider the 3\*50 iris flowers of known species as members of population 1, 2 and 3, respectively.

In the following, classification rules based on a Linear Discriminant Analysis (LDA) as well as a Quadratic Discriminant Analysis (QDA) will be constructed and performance evaluated assuming equal priors for each iris species and equal misclassification "costs"

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#### **Data preparation and plot of observations**

- Extract the size data for the three iris species observations from the data matrix
- Make a scatter plot of the bivariate observations from all three populations in the same figure using different colors for each population
- Calculate descriptive statistics for each population, i.e. sample mean vectors and sample covariance matrices

#### **Multivariate (bivariate) normal model check**

- For each of the three populations, perform a check of the MVN assumption that will be used in the following, that is, make scatter matrix plots and QQ-plots for sample Mahalanobis distances as in earlier problems
- Conclude (hopefully) that the MVN assumptions are reasonably verified

#### **LDA or QDA most appropriate ?**

- Perform the usual Bartlett test of a hypothesis of equal covariance matrices for the three populations
  - The low p-value for this test should indicate that the covariance matrices most probably are not equal, and thus QDA should be the most appropriate classification approach. Therefore it will be studied firstly in the following. For the sake of comparison between LDA and QDA, however, it will be followed by a study of LDA.
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### QDA case

- First, test if the three populations are significantly separated in terms of mean vectors, that is, make a one-way MANOVA test of a hypothesis of equal mean vectors. As usual this requires calculation of the sum-of-squares SSB and SSW. For the case present with three populations, an exact test-statistic with F-distribution is available using Wilks' Lambda (refer to textbook and earlier problems).
- Conclude that the populations indeed are significantly separated (that is, reject the hypothesis of equal mean vectors).

For the QDA case, one way of expressing the classification rule for equal priors, equal misclassification costs and observation  $\mathbf{x}$ , is done by defining the single-population QDA discriminant functions

$$d_i^{\text{QDA}}(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^T \mathbf{S}_i^{-1} \mathbf{x} + \bar{\mathbf{x}}_i^T \mathbf{S}_i^{-1} \mathbf{x} - \frac{1}{2}\log|\mathbf{S}_i| - \frac{1}{2}\bar{\mathbf{x}}_i^T \mathbf{S}_i^{-1} \bar{\mathbf{x}}_i, \quad i = 1, 2, 3$$

The classification rule for an observation  $\mathbf{x}$  is then simply

$$\hat{i}(\mathbf{x}) = \arg \max_{i=1,2,3} \{d_i^{\text{QDA}}(\mathbf{x})\}$$

that is, allocating the new observation to the population with highest value of the discriminant function

- Calculate the single-population discriminant functions  $d_1^{\text{QDA}}(\mathbf{x})$ ,  $d_2^{\text{QDA}}(\mathbf{x})$  and  $d_3^{\text{QDA}}(\mathbf{x})$  as quadratic functions of the variables  $x_1$  = sepal width [mm],  $x_2$  = petal width [mm].

An alternative way – better suited for plot of classification borders - of expressing the classification rule for equal priors, equal misclassification costs and observation  $\mathbf{x}$ , is done by defining the pairwise QDA discriminant functions

$$d_{ik}^{\text{QDA}}(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^T (\mathbf{S}_i^{-1} - \mathbf{S}_k^{-1}) \mathbf{x} + (\bar{\mathbf{x}}_i^T \mathbf{S}_i^{-1} - \bar{\mathbf{x}}_k^T \mathbf{S}_k^{-1}) \mathbf{x} - K_{ik}, \quad i, k \in \{1, 2, 3\}$$

where

$$K_{ik} = \frac{1}{2}\log \frac{|\mathbf{S}_i|}{|\mathbf{S}_k|} + \frac{1}{2}(\bar{\mathbf{x}}_i^T \mathbf{S}_i^{-1} \bar{\mathbf{x}}_i - \bar{\mathbf{x}}_k^T \mathbf{S}_k^{-1} \bar{\mathbf{x}}_k), \quad i, k \in \{1, 2, 3\}$$

The classification can then be expressed

$$\hat{i}(\mathbf{x}) = \begin{cases} 1 & d_{12}^{QDA}(\mathbf{x}) > 0 \wedge d_{13}^{QDA}(\mathbf{x}) > 0 \\ 2 & d_{21}^{QDA}(\mathbf{x}) > 0 \wedge d_{23}^{QDA}(\mathbf{x}) > 0 \\ 3 & d_{31}^{QDA}(\mathbf{x}) > 0 \wedge d_{32}^{QDA}(\mathbf{x}) > 0 \end{cases}$$

- Calculate the pairwise discriminant functions  $d_{12}^{QDA}(\mathbf{x})$ ,  $d_{13}^{QDA}(\mathbf{x})$  and  $d_{23}^{QDA}(\mathbf{x})$  as quadratic functions of the variables  $x_1 = \text{sepal width [mm]}$ ,  $x_2 = \text{petal width [mm]}$ .
- Make a new scatter plot of the bivariate observations from the three populations in the same figure and indicate the QDA classification borders given by  $d_{12}^{QDA}(\mathbf{x}) = 0$ ,  $d_{13}^{QDA}(\mathbf{x}) = 0$  and  $d_{23}^{QDA}(\mathbf{x}) = 0$

### Performance evaluation of QDA classification

- Calculate the confusion matrix for the known training data from the three populations using either the constructed decision rule for single-population QDA discriminant functions or the decision rule for pairwise QDA discriminant functions (or both) and verify the number of misclassifications with the figure just made
- Calculate the Apparent Error Rate, APER, for QDA using the training data
- Calculate the confusion matrix and the estimated actual error rate, AER, for training data using the cross-validation (hold-out) procedure
- Compare APER and AER for QDA

### Classification of new observations using QDA

Consider a new observation,  $\mathbf{x}_{01} = \begin{bmatrix} 2.6 \\ 1.5 \end{bmatrix}$ , for an iris flower of unknown species

- Calculate the discriminant functions  $d_i^{QDA}(\mathbf{x}_{01})$ ,  $i = 1, 2, 3$  and classify the new observation in terms of species
- Calculate the posterior probabilities for the new observation to be of each of the three species, respectively, which in the case of equal priors can be calculated as

$$P(i|\mathbf{x}_{01}) = \frac{\hat{f}_i(\mathbf{x}_{01})}{\hat{f}_1(\mathbf{x}_{01}) + \hat{f}_2(\mathbf{x}_{01}) + \hat{f}_3(\mathbf{x}_{01})}, \quad i = 1, 2, 3$$

where  $\hat{f}_i(\mathbf{x}_{01})$ ,  $i = 1, 2, 3$  are the “likelihoods”, that is the value of the MVN pdf’s for  $\mathbf{x} = \mathbf{x}_{01}$  using sample mean vectors and sample covariance matrices

- Classify the new observation again, now using the MAP (Maximum A posteriori) criterion, compare with above.
- Do the same for the new observations,  $\mathbf{x}_{02} = \begin{bmatrix} 2.6 \\ 1.6 \end{bmatrix}$  and  $\mathbf{x}_{03} = \begin{bmatrix} 3 \\ 0.75 \end{bmatrix}$

## LDA case

In spite of the result of the Bartlett test for equal population covariance matrices, for the sake of comparison, in the following the LDA case will be studied, thus anyway assuming that the covariance matrices are actually equal, implying that the three sample covariance matrices can be pooled to a common estimate.

For the LDA case the single-population discriminant functions – still assuming equal priors, equal misclassification costs and observation  $\mathbf{x}$  – reduces to a linear form

$$d_i^{\text{LDA}}(\mathbf{x}) = \bar{\mathbf{x}}_i^T \mathbf{S}_p^{-1} \mathbf{x} - \frac{1}{2} \bar{\mathbf{x}}_i^T \mathbf{S}_p^{-1} \bar{\mathbf{x}}_i, \quad i = 1, 2, 3$$

The classification rule for an observation  $\mathbf{x}$  is still

$$\hat{i}(\mathbf{x}) = \arg \max_{i=1,2,3} \{d_i^{\text{LDA}}(\mathbf{x})\}$$

that is, allocating the new observation to the population with highest value of the discriminant function

- Calculate the single-population discriminant functions  $d_1^{\text{LDA}}(\mathbf{x})$ ,  $d_2^{\text{LDA}}(\mathbf{x})$  and  $d_3^{\text{LDA}}(\mathbf{x})$  as linear functions of the variables  $x_1 = \text{sepal width [mm]}$ ,  $x_2 = \text{petal width [mm]}$ .

The alternative pairwise LDA discriminant functions simplifies to

$$d_{ik}^{\text{LDA}}(\mathbf{x}) = (\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_k)^T \mathbf{S}_p^{-1} \left[ \mathbf{x} - \frac{1}{2} (\bar{\mathbf{x}}_i + \bar{\mathbf{x}}_k) \right], \quad i, k \in \{1, 2, 3\}$$

with classification rule

$$\hat{i}(\mathbf{x}) = \begin{cases} 1 & d_{12}^{\text{LDA}}(\mathbf{x}) > 0 \wedge d_{13}^{\text{LDA}}(\mathbf{x}) > 0 \\ 2 & d_{21}^{\text{LDA}}(\mathbf{x}) > 0 \wedge d_{23}^{\text{LDA}}(\mathbf{x}) > 0 \\ 3 & d_{31}^{\text{LDA}}(\mathbf{x}) > 0 \wedge d_{32}^{\text{LDA}}(\mathbf{x}) > 0 \end{cases}$$

- Calculate the pairwise discriminant functions  $d_{12}^{\text{LDA}}(\mathbf{x})$ ,  $d_{13}^{\text{LDA}}(\mathbf{x})$  and  $d_{23}^{\text{LDA}}(\mathbf{x})$  as linear functions of the variables  $x_1 = \text{sepal width [mm]}$ ,  $x_2 = \text{petal width [mm]}$ .
- Make a new scatter plot of the bivariate observations from the three populations in the same figure and indicate the LDA classification borders (lines) given by  $d_{12}^{\text{LDA}}(\mathbf{x}) = 0$ ,  $d_{13}^{\text{LDA}}(\mathbf{x}) = 0$  and  $d_{23}^{\text{LDA}}(\mathbf{x}) = 0$

### **Performance evaluation of LDA classification and comparison with QDA**

- Calculate the confusion matrix for the known training data from the three populations using the constructed LDA classification (either single-population or pairwise discriminant functions) and verify the number of misclassifications with the figure just made
- Calculate the Apparent Error Rate, APER, using the training data
- Calculate the confusion matrix and the estimated actual error rate, AER, for training data using the cross-validation (hold-out) procedure
- Compare APER and AER for LDA
- Compare APER and AER using the QDA and LDA classification, respectively, and comment

### **Classification of the new observations using LDA and comparing with QDA**

Consider the same new observation as for QDA,  $\mathbf{x}_{01} = \begin{bmatrix} 2.6 \\ 1.5 \end{bmatrix}$ , for an iris flower of unknown species

- Calculate the discriminant functions  $d_i^{\text{LDA}}(\mathbf{x}_{01})$ ,  $i = 1, 2, 3$  and classify the new observation in terms of species
- Calculate the posterior probabilities for the new observation to be of each of the three species, respectively, which in the case of equal priors can be calculated as

$$P(i|\mathbf{x}_{01}) = \frac{\hat{f}_i(\mathbf{x}_{01})}{\hat{f}_1(\mathbf{x}_{01}) + \hat{f}_2(\mathbf{x}_{01}) + \hat{f}_3(\mathbf{x}_{01})}, \quad i = 1, 2, 3$$

where  $\hat{f}_i(\mathbf{x}_{01})$ ,  $i = 1, 2, 3$  are the “likelihoods”, that is the value of the MVN pdf’s for  $\mathbf{x} = \mathbf{x}_{01}$  using sample mean vectors and the pooled sample covariance matrix

- Classify the new observation again, now using the MAP (Maximum A posteriori) criterion, compare with above.
- Compare the classification using LDA with the previously classification using QDA and comment
- Do the same for the other two new observations,  $\mathbf{x}_{02} = \begin{bmatrix} 2.6 \\ 1.6 \end{bmatrix}$  and  $\mathbf{x}_{03} = \begin{bmatrix} 3 \\ 0.75 \end{bmatrix}$